

Python for Engineering: A Roadmap to Scientific Computing

Programming Foundations & Data Structures

Mutable vs. Immutable Data Structures

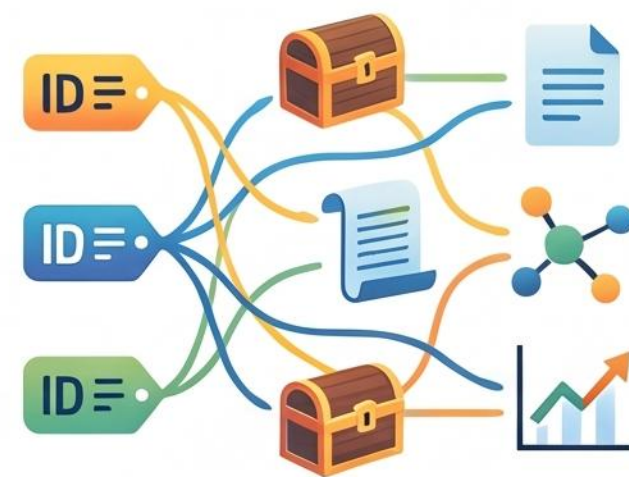


Lists are mutable for dynamic data



while Tuples are immutable to ensure data integrity

Key-Value Dictionary Mapping



Dictionaries enable efficient data retrieval and graph representation using unique key identifiers

Iterative Control Flow Selection

'for' loops



Use 'for' loops for known iterations

'while' loops



and 'while' loops for unknown convergence steps

Numerical Analysis & Matrix Operations

Vectorization with NumPy



NumPy arrays replace slow manual loops with high-performance element-wise and matrix operations

Numerical Solving with Gauss Elimination

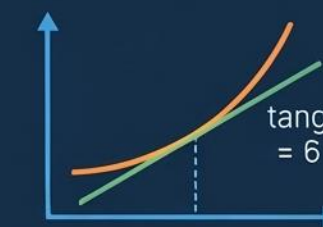
$$Ax = \begin{bmatrix} 1 & 2 & \dots & b \\ 0 & x & \dots & b \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & b \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ \vdots & \ddots & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Implementing pivoting and back-substitution to solve complex systems of linear equations ($Ax = b$)

Mathematical Modeling Tools



Taylor series



Numerical differentiation



Piecewise function

Native support for Taylor expansions, numerical differentiation, and piecewise function evaluation